

Lecture 19 (March 17)

Last class, we discussed how $L(m, n) \subset \mathbb{Y}$, consisting of Young diagrams that fit inside a $m \times n$ rectangle, is rank symmetric and unimodal. In 1980, Stanley proved that it is Sperner as well. Recall that having a symmetric chain decomposition would imply these three properties, so an open conjecture is that $L(m, n)$ always has a SCD. (This has been checked for specific values of m and n).

The second problem set is due on Friday.

Partition theory

We are studying the function $p(n)$, which is the number of partitions from n . Recall from investigating the limit of $\begin{bmatrix} n \\ k \end{bmatrix}_q$ that

$$\sum_{n \geq 0} p(n) \cdot q^n = \frac{1}{1-q} \cdot \frac{1}{1-q^2} \cdot \frac{1}{1-q^3} \cdots$$

(as proved by Euler in 1740). There is a more direct way to show this. To do this, we write a partition λ using different notation as $(1^{m_1} 2^{m_2} 3^{m_3} \dots)$ where m_i is the number of parts with size i in λ .

For example, $\lambda = (4, 4, 3, 1, 1, 1) \vdash 14$ can be written as $(1^3 2^0 3^1 4^2)$. Note that $|\lambda|$ will be $1 \cdot m_1 + 2 \cdot m_2 + 3 \cdot m_3 + \dots$

Then the generating function we want is

$$\begin{aligned} \sum_{\lambda} q^{|\lambda|} &= \sum_{m_1, m_2, m_3, \dots \geq 0} q^{1m_1 + 2m_2 + 3m_3 + \dots} \\ &= \left(\sum_{m_1 \geq 0} q^{m_1} \right) \left(\sum_{m_2 \geq 0} q^{2m_2} \right) \left(\sum_{m_3 \geq 0} q^{3m_3} \right) \cdots \\ &= \frac{1}{1-q} \cdot \frac{1}{1-q^2} \cdot \frac{1}{1-q^3} \cdots \end{aligned}$$

Since this infinite product converges, it is valid. (For any fixed n , only the first n terms can possibly affect the q^n coefficient.)

We can extend this idea to incorporate other constraints. Let's say you're interested in finding the ordinary generating function for the number of partitions of n with all parts $\leq k$. Then this just means $m_i = 0$ for all $i > k$, so the generating function is

$$\frac{1}{1-q} \cdot \frac{1}{1-q^2} \cdot \frac{1}{1-q^3} \cdots \frac{1}{1-q^k}.$$

Let $p_{\text{odd}}(n)$ denote the number of partitions of n with only odd-sized parts. (In other words, $m_i = 0$ for even i .)

Let $p_{\text{distinct}}(n)$ be the number of partitions of n with all distinct parts. (In other words, $m_i \leq 1$ for all i .)

Since we know what these means for each m_i , we can easily compute their generating functions:

$$\begin{aligned} \sum_{n \geq 0} p_{\text{odd}}(n) \cdot q^n &= \frac{1}{1-q} \cdot \frac{1}{1-q^3} \cdot \frac{1}{1-q^5} \cdots \\ \sum_{n \geq 0} p_{\text{distinct}}(n) \cdot q^n &= (1+q)(1+q^2)(1+q^3) \cdots \end{aligned}$$

Theorem 1 (Euler (1748)).

$p_{\text{odd}}(n) = p_{\text{distinct}}(n)$ for all n , so these generating functions are actually equal.

Proof. Let's rewrite our first generating function.

$$\begin{aligned} \frac{1}{1-q} \cdot \frac{1}{1-q^3} \cdot \frac{1}{1-q^5} \cdots &= \frac{(1-q^2)(1-q^4)(1-q^6) \cdots}{(1-q)(1-q^2)(1-q^3)(1-q^4) \cdots} \\ &= \frac{1-q^2}{1-q} \cdot \frac{1-q^4}{1-q^2} \cdot \frac{1-q^6}{1-q^3} \cdots \end{aligned}$$

Since $(1-q^{2n}) = (1-q^n)(1+q^n)$ (by difference of squares), this is indeed

$$(1+q)(1+q^2)(1+q^3) \cdots,$$

as desired.

Let's make a table of $p(n)$. From last lecture:

n	0	1	2	3	4	5	6	7	8	...
$p(n)$	1	1	2	3	5	7	11	15	22	...

After $p(n) = 4$, we may have a conjecture that $p(n)$ is the Fibonacci number, so that $p(n) = p(n-1) + p(n-2)$. Unfortunately, this recurrence breaks at $p(n) = 5$.

But maybe we can modify it somehow: we just need to subtract 1, so let's try $p(n) = p(n-1) + p(n-2) - p(n-5)$. This works for $n = 0 \dots 6$, but this would yield $p(7) = 16$. Maybe we should give up...

Euler didn't give up, though: it turns out $p(n-1) + p(n-2) - p(n-5) - p(n-7)$ works for $n < 12$. At that point, you need another correction: $+ p(n-12)$...

We can keep on going this way:

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + p(n-12) + p(n-15) - p(n-22) - p(n-26) \dots$$

Although you can do this for any sequence, there are some pretty surprising facts: the terms are relatively sparse, the coefficients are always ± 1 , positive and negative terms come in pairs, and the paired terms are closer than the gaps between the pairs...

Euler managed to explicitly figure out this recurrence, which makes computing $p(n)$ much faster than actually counting individual partitions.

What are these numbers, though? Let's find another way to obtain this recurrence.

We rewrite our generating function as

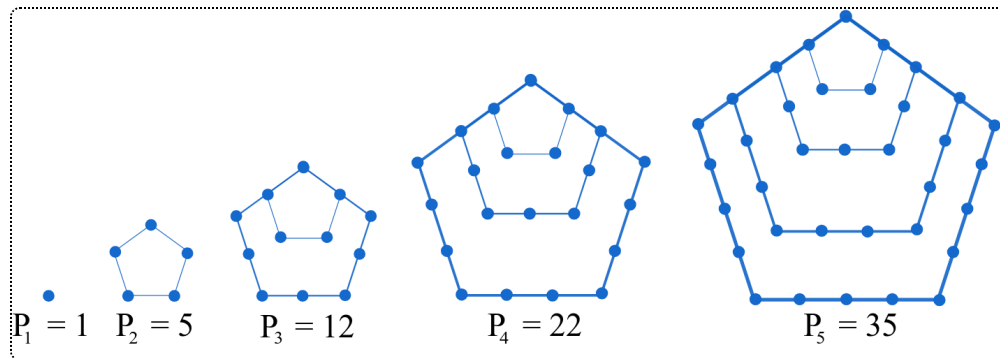
$$\left(\sum_{n \geq 0} p(n) \cdot q^n \right) \left(\prod_{k \geq 1} (1 - q^k) \right) = 1$$

Now expanding the power series $\prod (1 - q^k)$ exactly yields the recurrence:

$$\prod_{k \geq 1} (1 - q^k) = 1 - q^1 - q^2 + q^5 + q^7 - q^{12} - q^{15} + q^{22} + q^{26} - \dots$$

So our goal is really to understand this infinite product, and why it has our special properties.

Let's take every second number in our sequence, i.e. 1, 5, 12, 22, ... These numbers have a name: they are the **pentagonal numbers**, because they are related to pentagons. (Diagram below from math.net)



The explicit formula for these numbers is $\frac{k(3k-1)}{2}$. It's not hard to prove this.

Theorem 2 (Euler's pentagonal number theorem (1750)).

$$\prod_{k \geq 1} (1 - q^k) = \sum_{k \in \mathbb{Z}} (-1)^k q^{k(3k-1)/2}.$$

Note that in the RHS, k does not have to be positive. Negative values correspond to our secondary terms $-q^2, q^7, -q^{15}$, etc., and $k = 0$ corresponds to the 1 term.

Indeed, if $k' = -k$, $\frac{k(3k-1)}{2} = \frac{k'(3k'+1)}{2}$, which yields 2, 7, 15, 26, ... Sometimes people write the theorem with $k > 0$ and including both sequences explicitly, but this feels cleaner.

Euler actually conjectured this theorem in 1741, so it took him nearly ten years to prove it. We'll try to prove it in ten minutes!

Proof (due to Franklin). This uses involution, which can be a powerful proof method.

Recall that

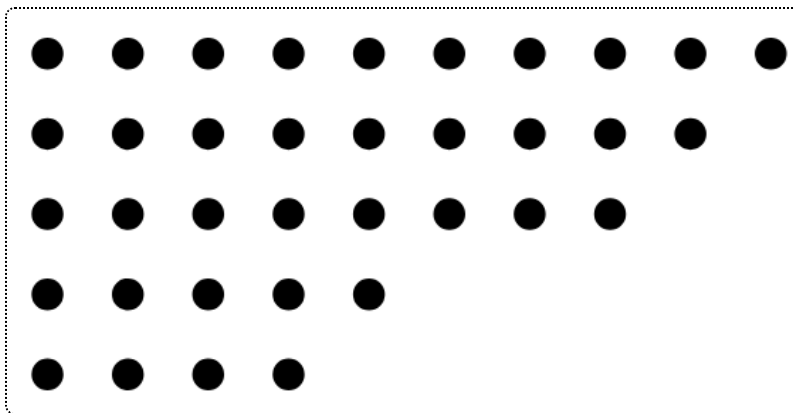
$$\prod_{k \geq 1} (1 + q^k) = \sum_{\lambda \text{ has distinct parts}} q^{|\lambda|}.$$

So if we change the $+$ to a $-$ in the RHS, we see that

$$\prod_{k \geq 1} (1 - q^k) = \sum_{\lambda \text{ has distinct parts}} (-1)^{\# \text{ parts in } \lambda} \cdot q^{|\lambda|}.$$

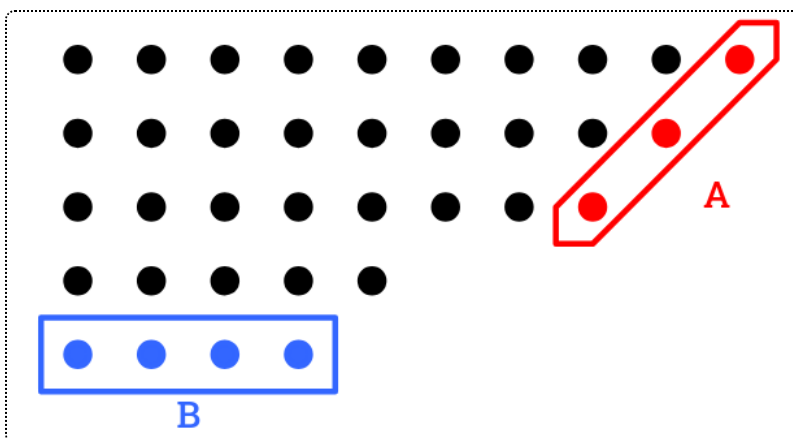
Somehow, we know that most of our partitions end up canceling out, so we want to construct an involution that pairs up partitions with the same size but a different number of parts (to pick which ones cancel with which).

Consider $\lambda = (10, 9, 8, 5, 4)$. We will draw a "Ferrers shape" (like a Young diagram but using dots instead of squares, since we don't need to put anything inside them).



We want to map this to some other diagram with the same number of dots, but such that the number of rows has changed parity.

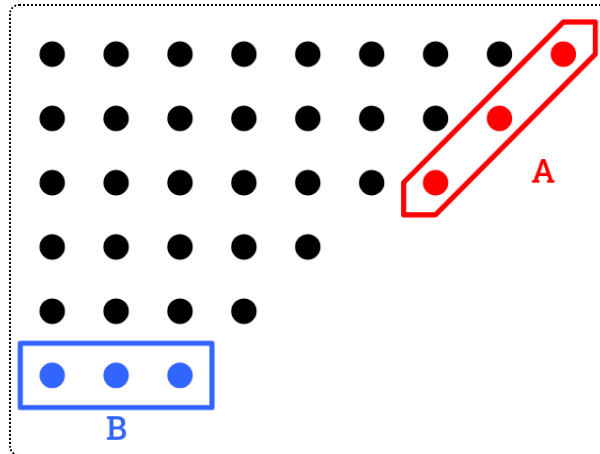
Consider two parts A and B : A is the set of dots in the rightmost "diagonal" of the diagram, while B is the bottom row.



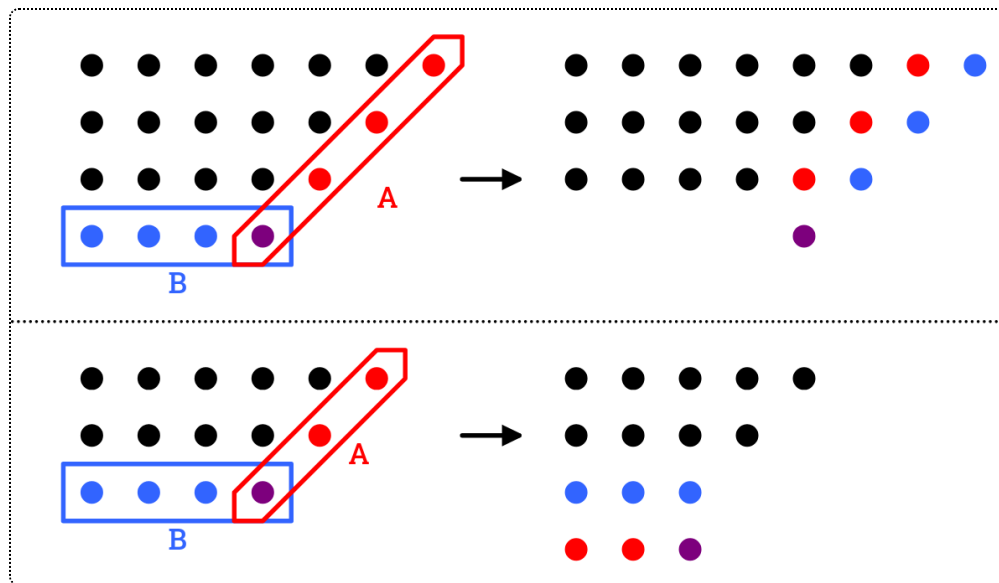
We have two cases.

1. If $|A| < |B|$, we move the dots of A down to make a new row.
2. If $|B| \geq |A|$, we do the opposite, moving the dots of B to the first $|B|$ rows.

So here, our λ is paired with the following $\tilde{\lambda} = (9, 8, 7, 5, 4, 3)$. The number of rows has changed by 1, so we pair up diagrams such that their row counts have different parity, as desired.

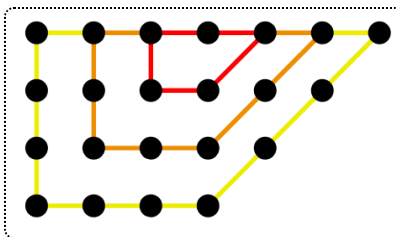


This works for most partitions, but there is one edge case where they don't work: when A and B intersect and either $|A| = |B|$ or $|A| = |B| - 1$. We can confirm that applying the operation here yields an invalid diagram.



As such, these Young diagrams never end up getting cancelled out in the sum, so their numbers of dots end up appearing in the nonzero terms. This is where the appearance of pentagonal numbers comes from: each of these diagrams is essentially just a collection of nested regular pentagons, albeit with two of the sides being collinear. The first case corresponds to $\frac{k(3k-1)}{2}$ and the second case

corresponds to $\frac{k(3k+1)}{2}$, and the signs alternate since the parity of the number of rows alternates.



This concludes the proof of the pentagonal number theorem.

Lecture 20 (March 19)

Please submit problem sets on Gradescope.

Jacobi triple product

There are other surprising theorems that are similar to Euler's pentagonal theorem. For example, cubing it yields:

Theorem 3 (Gauss).

$$\left(\prod_{k \geq 1} (1 - q^k) \right)^3 = \sum_{k \in \mathbb{Z}} (-1)^k k q^{k(k+1)/2}.$$

Perhaps most surprising is that squaring it yields a total mess. Maybe it would be interesting to see what exponents of $\prod(1 - q^k)$ result in nice expressions.

In order to understand that formula, let's take a look at this other product found by Gauss:

Theorem 4 (Gauss).

$$\prod_{n \geq 1} \frac{1 - q^n}{1 + q^n} = \sum_{k=-\infty}^{+\infty} (-1)^k q^{k^2}.$$

Taking infinite sums like the RHS requires some care. For example, consider this formal power series:

$$\begin{aligned}\sum_{n=-\infty}^{+\infty} x^n &= \sum_{n \geq 0} x^n + \sum_{n \leq -1} x^n \\ &= \sum_{n \geq 0} x^n + \frac{1}{x} \sum_{n \geq 0} \left(\frac{1}{x}\right)^n \\ &= \frac{1}{1-x} + \frac{1}{x} \cdot \frac{1}{1-\frac{1}{x}} = \frac{1}{1-x} + \frac{1}{x-1} = 0.\end{aligned}$$

This doesn't make any sense from an analytical point of view (in fact only one of the negative and positive series can ever converge) so be careful.

There is one formula that generalizes all three of these formulas (the two Gauss ones and the pentagonal number theorem), namely:

Theorem 5 (Jacobi Triple Product (1829)).

$$\prod_{n=1}^{\infty} \left((1 - x^{2n})(1 + x^{2n-1}y^2) \left(1 + \frac{x^{2n-1}}{y^2}\right) \right) = \sum_{k=-\infty}^{+\infty} x^{k^2} y^{2k}.$$

For convenience, we will substitute $x^2 = q$ and $y^2 = z \cdot q^{1/2}$. Then, we can rewrite this as

$$\prod_{n \geq 1} ((1 - q^n)(1 + zq^n)(1 + z^{-1}q^{n-1})) = \sum_{r=-\infty}^{+\infty} z^r q^{r(r+1)/2}.$$

Euler's pentagonal formula is the case where $q = t^3$ and $z = -t$.

Proof. First rewrite this as

$$\prod_{n \geq 1} (1 + zq^n) \prod_{n \geq 1} (1 + z^{-1}q^{n-1}) = \left(\sum_{r=-\infty}^{+\infty} z^r q^{r(r+1)/2} \right) \left(\prod_{n \geq 1} \frac{1}{1 - q^n} \right).$$

We want to understand these products and then find a combinatorial bijection between elements on either side.

What is the coefficient of z^a in the first product (denoted by the LHS of the following equation)? Since this is essentially our generating function for

partitions with distinct parts, this is

$$[z^a] \prod_{n \geq 1} (1 + zq^n) = \sum q^{|\mu|}$$

where μ is a partition with exactly a distinct parts.

Similarly,

$$[z^{-b}] \prod_{n \geq 1} (1 + z^{-1}q^{n-1}) = \sum q^{|\nu| - b}$$

where ν is a partition with exactly b distinct parts.

Finally, $\prod \frac{1}{1-q^n}$ is just $\sum q^{|\lambda|}$ over all partitions from before.

So, we want to form a bijection between tuples (μ, ν, a, b) on the LHS and pairs (λ, r) on the RHS.

- μ, ν are partitions with distinct parts
- a is the number of parts in μ ; b is the number of parts in ν
- λ is any partition, r is any integer
- We need to make sure that specific term matches, i.e.

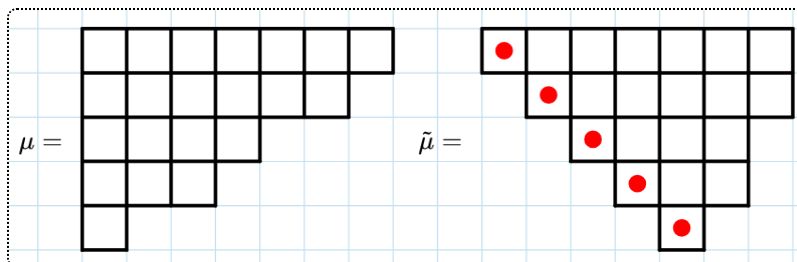
$$z^{a-b} q^{|\mu| + |\nu| - b} = z^r q^{r(r+1)/2 + |\lambda|}$$

- Therefore, $a - b = r$ and $|\mu| + |\nu| - b = \frac{r(r+1)}{2} + |\lambda|$.

Here's an example for the bijection; it should be pretty clear how to extend it.

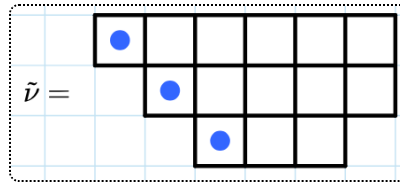
Suppose $\mu = (7, 6, 4, 3, 1)$ and $\nu = (6, 5, 3)$. Then $a = 5$ and $b = 3$.

We take μ 's Young diagram and "shift it" to yield $\tilde{\mu}$, in which each row has the same number of boxes but starts along the diagonal instead:



We can see how it looks like a staircase shape on the left and then can have any Young diagram shape on the right. We also draw a red dots at the start of rows.

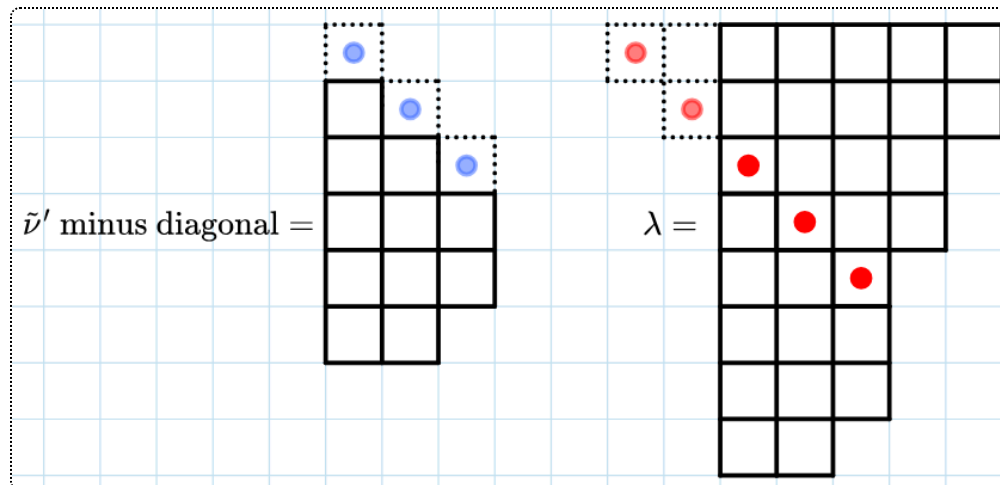
Similarly, here is $\tilde{\nu}$, with b blue dots.



We want to combine these two shifted Young diagrams to somehow yield λ . Here is the operation:

- Transpose $\tilde{\nu} \rightsquigarrow \tilde{\nu}'$
- Remove b diagonal boxes
- Glue it to $\tilde{\mu}$ at the last $\min(a, b)$ dots and chop off a staircase as necessary.

For example, we get $\lambda = (5, 5, 4, 4, 3, 3, 3, 2)$:



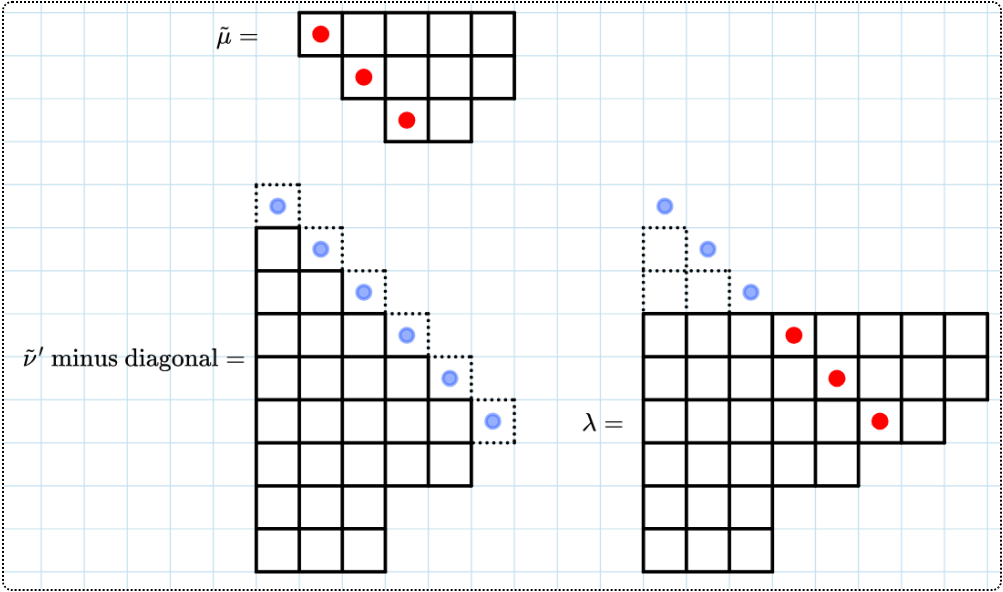
r corresponds to the side length of the chopped-off staircase, in this case 2.

The following properties that we needed are indeed verifiable based on this process:

- $a - b = r$
- The number of boxes $|\lambda|$ we end up with at the end is $|\mu| + |\nu| - b - \frac{r(r+1)}{2}$.

Here is an example for $a < b$ (which yields a negative value of r), for completeness: $\mu = (5, 4, 2)$, $\nu = (9, 8, 7, 4, 3, 1)$; $a = 3$, $b = 6$.

Now the size of the staircase is $b - a - 1$. In this case $r = a - b = -3$, and the area of the staircase is $\frac{(b-a-1)(b-a)}{2} = \frac{(-r-1)(-r)}{2} = \frac{r(r+1)}{2}$, as desired.



We also need to construct an inverse map $(\lambda, r) \rightarrow (\mu, \nu, a, b)$, but it's pretty straightforward. There are two cases ($r \geq 0$ and $r < 0$). In each, we attach a little staircase to the top-left of λ , draw the diagonal between $\tilde{\nu}'$ and $\tilde{\mu}$, and unshift to obtain μ and ν . All steps are reversible, as desired.